

Tien Soare (Nguyen)

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RESEARCH INTERESTS

Large Language Models (LLMs), program analysis, testing, debugging, and software engineering.
Focus on applying LLMs to real-world data-centric software engineering and reasoning tasks.

EDUCATION

Virginia Tech <i>Ph.D. in Computer Science</i>	Blacksburg, VA <i>Anticipated graduation: May 2028</i>
Virginia Tech <i>Master's in Computer Science</i>	Blacksburg, VA <i>Aug. 2023 – May 2026</i>
University of Saint Thomas <i>Bachelor in Computer Science, minor in Theology</i>	St. Paul, MN <i>Aug. 2020 – May 2022</i>
Century College <i>Associate's in Computer Science</i>	White Bear Lake, MN <i>Jan. 2018 – May 2020</i>

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML/CSS
ML & LLM Systems: vLLM (batch inference serving), quantization (FP8/INT8 evaluation), model evaluation and benchmarking, prompt design, PyTorch, HuggingFace Transformers
HPC & Infrastructure: Slurm, ARC/TinkerCliffs (NVIDIA A100), Docker, Conda, joblib, diskcache
Program Analysis: AST analysis, symbolic computation (SymPy), numerical stability, floating-point verification, pytest, JUnit
Developer Tools: Git, VS Code, PyCharm, Jupyter, Papermill, Cursor, LaTeX
Web: React, Flask, FastAPI, REST APIs, Material-UI

PUBLICATION

Are the Majority of Public Computational Notebooks Pathologically Non-Executable?

Tien Nguyen, Waris Gill, and Muhammad Ali Gulzar. *Accepted to MSR 2025. EMSE Special Invitee.*

- Analyzed 42,546 public Jupyter notebooks using static and dynamic approaches, introducing the first formal definitions of partial and pathological non-executability
- Classified notebooks into fully executable, partially executable, restorable, or pathologically non-executable categories based on runtime and dependency failures
- Found only 21.3% of notebooks are truly pathologically non-executable, much lower than the previously reported estimation of up to 76%
- Applied LLM-based generative techniques for code synthesis and repair, framing the task as structured generation and transformation, achieving full restoration for 5.4% and partial restoration for 12.8% of non-executable notebooks

Investigating and Remediating Pathological Non-Executability of Public Computational Notebooks

Tien Nguyen, Waris Gill, and Muhammad Ali Gulzar. *In review of EMSE. Extension of MSR 2025.*

- Built an automated testing tool to evaluate each notebook's executability, checking for required modules, external input files, and linear execution order
- Evaluated a curated dataset of 3,169 public notebooks, generating over 17,000 test cases and observing an average pass rate of 73.1%

Assessing Large Language Models for Stabilizing Numerical Expressions in Scientific Software

Tien Nguyen, Kirshanthan Sundararajah, and Muhammad Ali Gulzar. *In submission to ICSE 2027.*

- Designed the first evaluation of LLM reasoning for high-precision numerical computation across two tasks: *detection* (generating error-inducing inputs to expose instability in numerical expressions) and *stabilization* (rewriting expressions for numerical stability), verified by a two-stage semantic-equivalence check (structural AST analysis + numerical validation) to guarantee rewrites preserve semantics

- Curated a benchmark of 2,037 expressions and evaluated 4 LLMs over ~469K prompts and 3B tokens
- Found that while classical baselines stabilize more expressions overall (93.7% vs. 67.2%), LLMs uniquely fix 61.2% (79/129) of cases where baselines fail and beat baseline stability on 24% (489) of all expressions, positioning LLMs as a complementary tool for numerical robustness

Effects of Numeric Constant Representation and Quantization on LLM Numerical Reasoning

Tien Nguyen and Muhammad Ali Gulzar. In preparation.

- Built a SymPy-based rewriter that converts numeric constants to fraction, exact-rational, and symbolic forms while preserving precision and avoiding token collisions
- Evaluating representation effects on LLM numerical reasoning across multiple model families using vLLM batch inference on A100 GPUs
- Comparing full-precision and FP8 model pairs on numerical reasoning and code tasks, controlling for inference-engine confounds to isolate quantization effects

WORK EXPERIENCE

Graduate Research Assistant

August 2023 – Present

Virginia Tech

Blacksburg, VA

- Built and ran large-scale LLM evaluation pipelines on Virginia Tech’s ARC cluster, orchestrating vLLM inference across A100 GPUs under Slurm for 469K prompts and 3B tokens
- Developed static and dynamic analysis tooling for Python programs, including AST-based semantic-equivalence checking and symbolic verification with SymPy
- Collaborate with two faculty advisors and a research group across four concurrent projects, with regular written progress reporting

Information Security Student Coordinator

Feb. 2023 – May. 2023

University of Saint Thomas

St. Paul, MN

- Monitored and secured university systems and networks within the Security Operations Center (SOC)
- Conducted routine security audits to address about 20 system incidents per day
- Resolved about 5 IT service tickets per day using management tools for optimal efficiency
- Managed Azure Active Directory users, adhering to FERPA guidelines

Software Engineering Intern

Jun. 2021 – Aug. 2021

Target

Minneapolis, MN

- Built a full-stack web application that consolidated prepaid product masking codes from scattered sources into a single searchable tool, giving 100+ authorized users one reliable lookup point
- Developed responsive front-end components with React, JavaScript, Material UI, and Axios, streamlining search and retrieval for end users
- Implemented REST API endpoints on the back end using Java, Spring Boot, and Oracle Database to serve masking-code queries at scale
- Collaborated in an Agile team, using Jira for weekly sprint planning to deliver features iteratively

HONORS AND AWARDS

Grace Hopper Celebration Travel Scholarship <i>Virginia Tech, VA</i>	2026
Finalist (Top 6), Fowler Global Social Innovation Challenge	2021
Nancy Jo Hendrickson Legacy Scholarship <i>AAUW Northeast Metro Branch, MN</i>	Oct. 2020
Mathematics Scholarship <i>Century Foundation, MN</i>	Jul. 2020

TEACHING ACTIVITY

CS 1064 Introduction to Programming with Python, TA | *Virginia Tech*

- Fall 2024, Spring 2025, Summer 2026
- Led TA office hours with 200+ students per section

CS 1114 Introduction to Software Engineering with Java, TA | *Virginia Tech*

- Fall 2023, Fall 2025
- Led labs with 30+ students per section